Roadmap: ASA International Data Quest 2026 (WPP2024)

Data: UN World Population Prospects 2024 · Tool: Python notebook (.ipynb) · Goal: first place, with a standout Best Data Visualization

1. What the judges will score

Each team is scored 0–100 by three independent judges from outside the participating universities. A judge reviews up to eight teams, so your entry must be understood quickly and stand out.

Category	Weight	Sub-criteria (points within the category)
Best Data Visualization	35%	Clarity (35) · Creativity (30) · Effectiveness of charts, graphics and storytelling (35)
Best Consideration of Ethics	30%	Awareness of ethical issues in data analysis (35) · Fairness, transparency, privacy (30) · Avoiding misleading interpretations (35)
Best Statistical Insight	35%	Depth and originality of analysis (35) · Appropriate methodology (35) · Contribution to meaningful conclusions (30)

How to read this

- The three categories are almost equal. A beautiful but shallow entry loses 65%. Ethics is worth 30%, and most teams will treat it as an afterthought, so doing it well is the cheapest way to get ahead.
- Judges read slides and watch a video. Everything you want credit for must be visible there, not only in the notebook.
- Deck rule from the handbook: a cover slide, a references slide, and at most 4 content slides. Slides should stress visualizations, insights and ethical considerations rather than dense text. A video accompanies the deck.

Confirmed vs still to verify. Confirmed: the scoring table, the deck rule, and the data guide. Not yet confirmed (check the Timeline, Rules and Procedures, Language, Awards and Video Submission Guide sections): deadline and time zone, team size and eligibility, language, video length and format, submission channel, whether code must be submitted, and whether there are separate category awards in addition to an overall winner.

2. The roadmap, step by step

Step 0 — Lock down the rules first

The competition runs in October and today is 9 October, so confirm the deadline before anything else. Write down: deadline and time zone, team size, language, video length and format, deck format and file limits, how and where to submit, and whether an appendix or code is allowed. Ask your country representative about anything unclear. Plan to finish with a buffer, not at the deadline.

Step 1 — Environment and reproducibility

- Create one project folder with `data/`, `notebooks/`, `figures/`, `slides/`. Put all data files in `data/`.
- Install and pin: pandas, numpy, scipy, scikit-learn, statsmodels, matplotlib, seaborn, plotly, kaleido (static export of Plotly), and optionally tslearn (time-series clustering). Save `requirements.txt`.
- Fix random seeds everywhere. The notebook must run top to bottom on a fresh kernel.
- Start an **analysis log** (a markdown cell or file): every decision, every dead end, every test you ran. You will need it for the ethics criteria (a priori vs post hoc, multiple comparisons).

Step 2 — Data audit (get the foundations right)

1. **Load** the Medium file with `encoding='utf-8-sig'` and `compression='gzip'`, plus the notes file and the supplement. You still need to add `WPP2024_Demographic_Indicators_Medium_csv.gz` to your folder; it was not in the files uploaded here.
2. **Countries only.** `ISO3_code` is filled only for real countries. Filter on it before any cross-country comparison or sum, otherwise population is double-counted. Check `LocTypeName` value counts to understand the aggregates, and keep region and income-group labels aside for later comparison (never sum them with countries).
3. **Sanity check** that country totals match the World row:

```
chk = countries.groupby('Time')['TPopulation1July'].sum()
world = medium[medium['Location'] == 'World'].set_index('Time')
['TPopulation1July']
print(((chk - world) / world).abs().describe())   # should be very small
```

4. **Merge the supplement** on `ISO3_code` and `Time` (left join; Holy See has no rows there). Verify `BirthsMale + BirthsFemale` matches `Births` within 0.001 and that `LE60` lies between the male and female values. For sex ratios use `SRB` from the

Medium file, not the supplement, because the supplement's births are rounded.

5. **Flag estimates vs projections:** years up to 2023 are estimates, 2024 onward are projections. Create `is_projection` and carry it into every figure.
6. **Missing values:** build a table of missingness by indicator and country, decide how to treat each case, and write the decision down. Methods slides and the ethics criteria both reward explicit handling of missing data.
7. **Units:** take them from the notes file (thousands, per 1,000, years, and so on). Label every axis with units.
8. **Write a one-paragraph data card:** source (UN DESA Population Division, WPP 2024), license (CC BY 3.0 IGO), citation, unit of analysis (country-year), and known caveats (aggregated national data, modeled estimates, the guide's erratum about LE60 and births by sex, supplement rounding).

Step 3 — Pick a sharp question and write hypotheses before looking

The guide's suggestions are open-ended, so the winner is the team with a *specific* question and a clear answer. Recommended headline question:

> **How many distinct demographic pathways have countries followed since 1950, which countries are on diverging paths, and how robust are those groups to modeling choices and projection uncertainty?**

Why this one: it matches two of the guide's own prompts (combining indicators into a profile, and grouping countries by how indicators moved together over time), it lets you show real statistical depth, and it has a natural ethics angle (how to label and compare countries fairly).

Write your hypotheses now and date them, for example:

- H1: pathways are more varied than a simple rich/poor or regional split.
- H2: in the Medium projection the groups move closer together by 2100, but the spread under High and Low variants stays wide.
- H3: a minority of countries change group depending on modeling choices.

Anything you add after seeing results is labeled **post hoc** on your slides. Backup question if clustering shows no real structure: *how do aging indicators (median age, LE60, TFR) diverge across countries?* Originality ideas that use the supplement: separate longevity gains at birth from gains after age 60 (LEx vs LE60), and compute a **transition timing** feature (the year TFR crossed a threshold such as 4 and 2.5).

Step 4 — Exploratory analysis (clearly labeled exploratory)

- Look at distributions and trends of TFR, LEx, LE60, MedianAgePop, PopGrowthRate, CNMR for all countries; spot outliers (conflict years, migration spikes, very small island populations).
- Check correlations and redundancy (for example LEx, LE60, IMR and Q5 move together) and decide which indicators enter the analysis and why.
- Record every look in the analysis log and mark these results *exploratory*. Do not present an exploratory pattern as a confirmed finding.

Step 5 — Core statistical analysis (35% of the score)

5.1 Feature design. Build one feature set per country from 1950–2023 estimates only:

- Trajectory summary features: level in 1950 and 2023, speed of change, timing of the main transition, volatility.
- A second version using the raw yearly or 5-year trajectories (as a robustness check).
- Standardize each feature (z-scores or robust scaling). Decide how to treat very small populations and justify it (keep and flag, or exclude and show a sensitivity run).

5.2 Dimension reduction. Run PCA; report variance explained and loadings and interpret the components in plain language.

5.3 Clustering. Compare at least k-means, Ward hierarchical clustering and a Gaussian mixture model (and optionally DTW-based clustering for trajectories). Choose the number of groups with several criteria (silhouette, Calinski–Harabasz, gap statistic, BIC) and report all of them, including where they disagree.

5.4 Validate the clusters. This is where most entries stop too early.

```
from sklearn.cluster import KMeans
from sklearn.metrics import adjusted_rand_score
import numpy as np
rng = np.random.default_rng(42)
base = KMeans(k, n_init=50, random_state=42).fit_predict(X)
aris = []
for b in range(500):
    idx = rng.choice(len(X), size=int(0.8 * len(X)), replace=False)
    lab = KMeans(k, n_init=20, random_state=b).fit_predict(X[idx])
    aris.append(adjusted_rand_score(base[idx], lab))
print(np.mean(aris), np.percentile(aris, [2.5, 97.5]))
```

- Stability: bootstrap or subsample countries and report the adjusted Rand index; compute a per-country membership confidence (bootstrap frequency or mixture posterior) so borderline countries are visible.

- External validity: compare clusters with UN regions and income groups (Cramér's V, adjusted Rand index) to show what the clusters add beyond geography.
- Avoid circular testing: do not claim clusters differ on the same variables used to create them. Validate on held-out indicators such as PopDensity or net migration.

5.5 Go beyond static groups. Add one analysis that others will not have, for example: sigma-convergence (cross-country dispersion of TFR and LEx over time), or cluster transitions across snapshot years (1950, 1980, 2010, 2040, 2070, 2100) shown as a flow diagram, with the projection years marked.

5.6 Multiple comparisons. If you run many tests, say so and adjust (Holm or Benjamini–Hochberg).

Step 6 — Uncertainty and robustness

- Use the optional **OtherVariants** file. It is large (about 660,000 rows), so load only what you need (`usecols`, filter by `Variant`).
- For each country compare 2050 and 2100 values of population, TFR and median age under Medium, High, Low and Constant fertility, and the 80%/95% probability intervals.
- Re-assign countries to groups using the High and Low projections and report the share that change group.
- Run a sensitivity grid: feature set × algorithm × k × scaling × inclusion of small countries. Summarize it in one compact figure.
- If a conclusion is fragile, say so. Honest fragility is rewarded under both the methodology and ethics criteria.

Step 7 — Turn results into conclusions

Write 3–4 findings. For each: the claim, the evidence, the size of the effect, the uncertainty, and one caveat. Then add the so-what: what a reader should understand or do differently. Use calibrated language and avoid causal wording (the data are observational and country-level, so beware the ecological fallacy). Do not give policy prescriptions the data cannot support.

Step 8 — Build the ethics layer into the work (30% of the score)

Do not write an ethics paragraph at the end. Show ethical practice in the process and in the visuals. Use the ASA *Ethical Guidelines for Statistical Practice* (2022) as your framework and cite it.

ASA guideline	What to do in this project
B1 Data sources, fitness for use, missing data	Data card; missingness table; state that many country values are modeled estimates and that data quality differs between countries
B2 Assumptions and limitations	One visible limitations box on the deck; projections are one plausible scenario, not a forecast
B3 A priori vs post hoc; multiple comparisons	Dated hypotheses; label post hoc findings; report adjustments
A2, F4 Valid and reproducible methods	Seeds, pinned packages, clean notebook, shareable code
A4, C2 Do not selectively interpret or mislead	Show all values of k you tried and results that did not support your hypotheses
D6, D11 Impact on groups and framing	Neutral cluster names that describe patterns (never ‘backward’ or ‘advanced’); no league tables or country rankings; explain that clusters are descriptive, not judgments
D4, D7, D9 Privacy and confidentiality	State plainly that the data are aggregated national indicators with no individual-level records, so re-identification risk is minimal; still note small-population noise
B7 Variation across groups	Show results by region and income group and by sex where available (LE60 male vs female)
D10 Provenance	UN WPP 2024 citation, CC BY 3.0 IGO, and the supplement's note on rounding and the missing Holy See

Ethical risks specific to this dataset to mention explicitly: projections read as certainties, differences in data quality between countries, the ecological fallacy (national averages hide within-country inequality), political and territorial labels, and stigmatizing labels for groups of countries.

Step 9 — Visualization design (35% of the score)

Principles, tied to the three sub-criteria:

- **Clarity (35):** one message per chart; titles that state the finding; direct labels instead of legends where possible; consistent cluster colors on every figure; large fonts that stay readable in a video; units on axes; source footnote.
- **Creativity (30):** choose one or two signature visuals (below) and make them look distinctive and professional.
- **Effectiveness and storytelling (35):** match each chart type to its question (trends → lines, geography → map, relationships → scatter, uncertainty → bands) and order the figures as a story: question → pattern → how sure are we → what it means.

Design rules that also protect your ethics score: zero baselines for bars; the same axes across small multiples; no dual axes that invite false correlation; a vertical line or lighter style at the 2023/2024 boundary to separate estimates from projections; uncertainty bands wherever you show the future; a colorblind-safe palette (Okabe–Ito) and a grayscale check.

Figure candidates:

1. **World map of pathways** (Plotly choropleth by `ISO3_code`, so no shapefile needed).
2. **Path plot in the TFR–life expectancy plane**, with one line per country coloured by group, ideally animated by year in the video.
3. **Cluster transition flow diagram** (Sankey) across snapshot years.
4. **Uncertainty fan chart** (Medium vs High/Low and probability intervals) for representative countries or groups.
5. **Stability heatmap** of cluster robustness with per-country confidence.
6. **PCA biplot** to explain what the groups mean.

Export static images at 300 dpi or as SVG, sized for 16:9 slides.

Step 10 — The slide deck (cover + 4 content slides + references)

Slide	Purpose	Hero visual	Ethics element
Cover	Title, the question in one line, team	—	—
Content 1	The question and the big picture	World map of pathways	Data source and caveat line
Content 2	How the pathways differ	Path plot / trajectories with the projection boundary marked	Neutral naming of groups
Content 3	How sure are we	Stability heatmap plus uncertainty fan chart	Limits and sensitivity stated openly

Slide	Purpose	Hero visual	Ethics element
Content 4	So what, and responsible use	Key findings with effect sizes	Ethics summary: scenarios not forecasts, data quality, no rankings
References	Citations	—	UN WPP 2024, ASA ethical guidelines, tools

Put a one-line **method strip** (features → PCA → clustering → validation) on slide 2 or 3, since methodology is scored. Keep text minimal. Check whether the rules allow an appendix or code link for details.

Step 11 — The video

Once you know the required length, write the script to fit it (about 130 words per minute). Suggested flow: a hook (10 seconds), the question, data and ethics note, method in one breath, three findings, uncertainty, conclusion. Walk through the visuals with a pointer or animation so the viewer's eye is guided. Use clear audio, add subtitles if allowed, rehearse at least twice, and keep to the language rule.

Step 12 — Quality check and submission

- Re-run the notebook on a fresh kernel. Every number on the slides must match the notebook.
- Every chart has a takeaway title, units, source and uncertainty where relevant.
- Ask someone outside your team to read the deck for 2 minutes and tell you the main message; if they cannot, simplify.
- Check file names, sizes and formats; submit before the deadline with time to spare; keep a copy of everything.

3. Score yourself like a judge

Criterion	What a top entry looks like
Clarity	A stranger understands every chart in under 10 seconds
Creativity	One or two memorable visuals that are still honest
Effectiveness and storytelling	The figures tell one story in order, and each chart type fits its question

Criterion	What a top entry looks like
Awareness of ethical issues	Dataset-specific risks named, not generic statements
Fairness, transparency, privacy	Neutral labels, open assumptions, privacy addressed, reproducible code
Avoiding misleading interpretations	Uncertainty shown, estimates vs projections separated, no causal claims
Depth and originality	At least one analysis beyond standard clustering (validation, convergence, transitions)
Appropriate methodology	Choices justified; alternatives compared; multiple criteria for k
Meaningful conclusions	Findings with effect sizes, caveats and a clear so-what

4. Mistakes that lose points

- Summing or comparing countries together with regional or income aggregates.
- Mixing estimates and projections without marking the boundary.
- Presenting the Medium projection as a prediction instead of one scenario.
- Testing differences between clusters on the same variables used to build them.
- Choosing k because it looks nice, without showing the alternatives.
- Country rankings or loaded names for groups.
- Dense slides, tiny fonts, or ten charts where four would do.
- An ethics section added only at the end.
- Overly complex methods that you cannot explain in the video.

5. Open items to confirm

- [ ] Deadline and time zone
- [ ] Team size and eligibility
- [ ] Required language
- [ ] Video length, format and file size
- [ ] Deck format and whether an appendix is allowed
- [ ] Whether code or the notebook must be submitted
- [ ] Award structure (overall vs per category)

- [ ] Main Medium data file added to your folder